

Ecstatic over extactic

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1 Abstract

In a second course on calculus, a common topic is vector geometry, and in particular the properties of vector curves $\vec{r}(t)$. At a particular point $t = t_0$, the derivatives of $\vec{r}(t)$ are evaluated and shown to represent the tangent and the normal to the curve, as well as the local curvature of the curve. Often mentioned in passing is the osculating circle to the curve at t_0 .

These properties were studied by Cayley in several publications[2], including one in 1859. He wrote about these properties in an amusing way. He defined a tangent to the curve as a line that passes through 2 consecutive points on the curve and the osculating circle as the circle that passes through 3 consecutive points. That language might already be a challenge for students, but he continued. He threw out the term, without proper definition, *sextactic*. Since Cayley practiced as a lawyer for 24 years while he tried to obtain a professorship in mathematics, it is natural to assume that he thought about latin terms and phrases a great deal. His term *sextactic* then becomes obvious (to latin readers) as *sex*=6 and *tactic*=touching, from latin *tactilus*, which gives us English words such as *tactile*. So sextactic is presumably a point of contact between 2 curves which touch each other at 6 consecutive points.

From this, various authors have invented the word *extactic*, but there are problems with this. First, there is no agreed definition on what it means, and secondly, the authors cannot spell their new word, and one sees *extatic* in the titles of papers [1]. Hence the present title takes advantage of the potential pun.

The talk will use Maple to automate some of the calculations used to study the curves. This should provide some useful guidance for those teaching a second course on calculus, and wish to automate the routine calculations appearing in the textbooks. We shall also tackle Cayley’s colourful definitions of what tangents and circles.

Keywords

Vector curves, Osculating circle, undergraduate education

References

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- [2] Cayley, A.: On the conic of five-pointic contact at any point of a plane curve. Philosophical Transactions of the Royal Society of London **149**, 371–400 (1859)